CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Tinkercad 2: Holes and group

Goal: I can use a hole shape and group to make a keychain tag with a working hole.

35 minutes | 7 steps | Computer Science and Programming, Digital Literacy | Playbook A meeting 25

Ask | Plan | Build | Test | Improve

You are done when

Tell a partner how you made the hole go all the way through.

You will use

- Cart laptops, charged
- Tinkercad Classroom class code
- Projected teacher demo
- Card, hole punch and grid paper for the no-device path

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Sign in — 4 min

- Get your laptop.
- Open Tinkercad with the class code.

2 Step 2: Recap — 1 min

- Show a partner how to lift a shape.

3 Step 3: Watch a hole — 4 min

- Watch a cylinder turn into a hole.
- Watch Group cut it out.

4 Step 4: Make a tag — 14 min

- Make a box for your keychain tag.
- Set a cylinder to Hole and push it through a corner.
- Select both and choose Group.

5 Step 5: Check the hole — 5 min

- Can you see through the hole?
- If not, push it all the way through.
- Show a partner the hole from underneath.

6 Step 6: Save and shut — 4 min

- Check that your work saved.
- Close the lid with two hands.
- Put the laptop back in its slot and plug it in.

7 Step 7: Tell how — 3 min

- Tell a partner how you made the hole go all the way through.

What to turn in

A keychain tag with a hole you can see through.

No computer? Another way to do it

Draw the tag to size, punch a clean hole and join two card layers.

Finished early? Next challenge

Add a raised letter and group it with the tag.

Stay safe

Nicknames or initials only. Laptops stay on the table.

Grade targets

Grade 5: Make the tag and the working hole.
Grade 6: Add a raised letter grouped with the tag.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.